Unit-1

Introduction

Artificial Intelligence (AI)

Artificial Intelligence is a branch of computer science that focuses on creating machines and software that can think, learn, and act like humans. It enables computers to perform tasks that normally require human intelligence, such as learning from experience, reasoning, problem-solving, understanding language, and making decisions.

AI Problems

An AI problem is any task where a machine must exhibit intelligent behavior such as reasoning, learning, planning, or decision-making to achieve a goal under given constraints. These problems usually involve uncertainty, large state spaces, and complex decision processes.

The goal is to make machines solve problems like humans even when:

- Information is incomplete
- Search space is large
- Environment is uncertain

Characteristics of AI Problems

1. Large Search Space

AI problems usually involve a very large number of possible states and actions, making it impossible to check every option manually. For example, in chess or route planning, the AI must intelligently search through many possibilities to find a good solution.

2. Uncertainty

In many AI problems, the available information may be incomplete, noisy, or changing over time, so outcomes cannot be predicted with full certainty. For instance, weather forecasting systems must work with uncertain data and still provide useful predictions.

3. Multiple Possible Solutions

AI problems often do not have a single correct answer; instead, there may be many acceptable solutions. The goal of AI is to find the best or most efficient solution among several options, such as selecting the shortest or safest path.

4. Requires Knowledge

AI systems depend on stored knowledge such as facts, rules, and relationships to solve problems effectively. For example, a medical diagnosis system uses medical knowledge to identify diseases based on symptoms.

5. Needs Reasoning or Learning

AI uses reasoning techniques like logic to derive new information from existing knowledge, or learning methods to identify patterns from data. This helps the system improve its performance over time and make better decisions.

6. Complex Representations

AI problems often require complex ways of representing information, such as graphs, logical rules, or probability models. These representations help the AI understand and handle real-world complexity efficiently.

Types of AI Problems

1. Search Problems

Search problems involve finding a sequence of actions that takes an agent from an initial state to a goal state. The agent explores different possible states and chooses a path that successfully reaches the goal, such as finding a route or solving a puzzle.

Examples:

- Pathfinding (GPS)
- 8-puzzle, chess
- Robot navigation

2. Constraint Satisfaction Problems (CSP)

In constraint satisfaction problems, the solution is found by assigning values to variables while satisfying all given constraints. The focus is not on the path but on finding a valid combination that meets all rules.

Examples:

- Sudoku
- Map coloring
- College timetable scheduling

3. Planning Problems

Planning problems require the agent to decide a step-by-step sequence of actions to achieve a specific goal. The agent must consider the order of actions and their effects before execution.

Examples:

- Robot task planning
- Logistics / delivery routes
- Manufacturing plans

4. Reasoning Problems

Reasoning problems use logic and inference to derive new information from known facts. The agent analyzes rules and facts to reach conclusions or make decisions.

Examples:

- Medical diagnosis
- Expert systems
- Legal reasoning

5. Learning Problems

Learning problems involve improving performance by learning patterns from data or past experience. The agent adjusts its behavior based on examples or feedback.

Examples:

- Spam email detection
- Neural network training
- Predicting house prices

6. Perception Problems

Perception problems focus on interpreting raw sensory input from the environment. The agent converts signals such as images or sounds into meaningful information.

Examples:

- Speech recognition
- Image classification
- Sensor fusion in robots

7. Natural Language Processing (NLP)

NLP problems deal with understanding and generating human language in written or spoken form. The agent processes language to communicate or extract meaning from text.

Examples:

- Chatbots
- Machine translation
- Sentiment analysis

8. Multi-Agent Problems

Multi-agent problems involve multiple intelligent agents interacting within the same environment. Each agent's actions can influence others, requiring coordination or competition.

Examples:

- Autonomous vehicle coordination
- Game-theoretic tasks
- Multi-robot systems

Steps in Solving an AI Problem

Step 1: Problem Definition

In this step, the AI problem is clearly defined by specifying the initial state, the goal state, the possible actions, and the constraints. A clear problem definition helps the AI understand what needs to be solved and under what conditions.

Step 2: Representation

The problem is represented in a suitable form such as graphs, logic, rules, features, or probability models. Proper representation makes it easier for the AI system to process information and apply problem-solving methods.

Step 3: Search / Reasoning

The AI uses search or reasoning techniques like BFS, DFS, A*, constraint satisfaction, or logical inference to explore possible solutions. These methods help the system decide the best path or action to reach the goal.

Step 4: Learning / Optimization

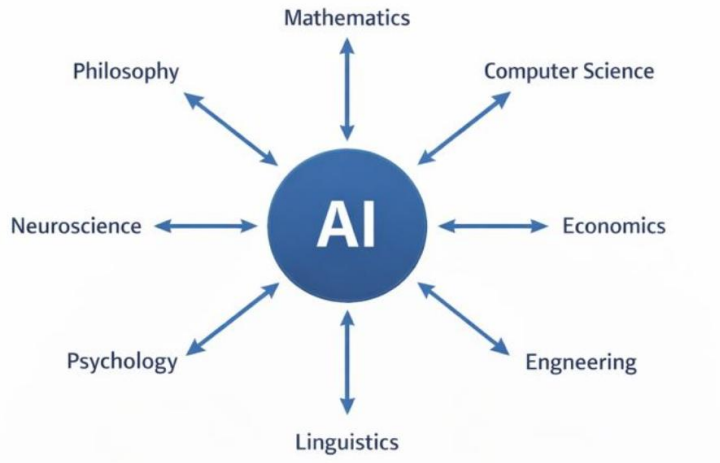
In this step, the AI improves its performance by learning from data or past experience if required. Machine learning techniques are used to optimize decisions and produce better results over time.

Step 5: Execution & Feedback

The selected solution is executed in the real or simulated environment. Feedback is then collected and used to update the knowledge base or model for future improvement.

Foundation of Artificial Intelligence

The foundation of Artificial Intelligence refers to the basic disciplines and concepts that help in building intelligent systems. AI is not an independent subject but is developed using ideas from many fields that explain how machines can think, learn, and make decisions like humans.



1. Philosophy (Logic & Reasoning)

Philosophy contributes ideas related to logic, reasoning, and rational thinking. It helps AI systems represent knowledge and make logical decisions based on facts, goals, and beliefs, which is important for intelligent behavior.

2. Mathematics

Mathematics provides the formal tools required to handle calculations, uncertainty, and optimization in AI. Concepts like probability, statistics, linear algebra, and logic help AI systems analyze data, learn patterns, and find optimal solutions.

3. Computer Science

Computer Science provides algorithms, data structures, and programming techniques needed to implement AI systems. It helps in designing efficient search methods, handling large amounts of data, and developing intelligent software and applications.

4. Neuroscience

Neuroscience studies how the human brain works and processes information. AI takes inspiration from the structure of neurons and learning processes of the brain to develop artificial neural networks and learning models.

5. Psychology & Cognitive Science

Psychology and cognitive science explain how humans think, learn, solve problems, and perceive the world. These ideas are used in AI to design systems that can learn from experience and behave intelligently like humans.

6. Linguistics

Linguistics deals with the structure and meaning of language. It helps AI systems understand, process, and generate human language, which is essential for applications like speech recognition, chatbots, and machine translation.

7. Engineering

Engineering contributes practical knowledge for building physical AI systems such as robots. It helps in designing sensors, control systems, and hardware that allow AI systems to sense the environment and perform actions.

8. Economics (Decision Theory & Game Theory)

Economics provides methods for making rational decisions under different conditions. Concepts like decision theory and game theory help AI systems choose optimal actions, especially in situations involving multiple agents.

History and Generations of Artificial Intelligence

1. The Generation of Artificial Intelligence

In the beginning, scientists started wondering whether machines can think like humans. This idea came from logic, mathematics, and early brain studies. Alan Turing proposed that if a machine can behave like a human in conversation, it can be considered intelligent. This stage created the theoretical foundation of AI.

2. The Birth of Artificial Intelligence

Artificial Intelligence officially started in 1956 at the Dartmouth Conference. The term *Artificial Intelligence* was introduced here. Researchers believed that human intelligence could be described using rules and implemented in computers. This event marked AI as a separate research field.

3. Early Enthusiasm

After AI was born, researchers became very optimistic. Early programs solved puzzles, played games, and proved mathematical theorems. This success created excitement, and many believed machines would soon match human intelligence. Funding and research increased rapidly during this phase.

4. Great Expectations

Because early systems worked well in small problems, expectations became very high. People believed AI would understand language, reason like humans, and solve real-world problems easily. However, these expectations were unrealistic because real environments are complex and uncertain.

5. A Dose of Reality

When AI systems failed in real-world situations, researchers realized the limitations. Computers were slow, data was limited, and systems could not handle uncertainty. Funding was reduced, leading to the first AI winter. This phase taught that intelligence is much harder than expected.

6. Knowledge-Based Systems

AI recovered by focusing on expert knowledge instead of general intelligence. Expert systems used rules written by human experts to solve specific problems like medical diagnosis. These systems showed that knowledge is central to intelligence, which became a key idea in AI.

7. AI Becomes an Industry

Because expert systems worked well, AI entered real industries. Companies used AI in medicine, manufacturing, and business decision systems. AI moved from research labs to practical applications. However, these systems were costly and difficult to maintain.

8. The Return of Neural Networks

Later, AI researchers returned to neural networks, inspired by the human brain. With better computers and learning algorithms, machines could learn from data instead of being manually programmed. This marked the shift toward machine learning.

9. AI Becomes a Science

AI then became more scientific and disciplined. Researchers started using mathematical models, experiments, and performance evaluation. According to AIMA, AI focused on building rational systems that act correctly based on evidence, making AI a true science.

10. The Emergence of Intelligent Agents

Modern AI views systems as intelligent agents that sense the environment and take actions to achieve goals. This approach unifies all AI concepts like search, learning, reasoning, and decision-making. Intelligent agents represent the modern view of AI.

Agent

An agent is anything that can observe its environment and take actions. In simple words, an agent is something that senses, decides, and acts.

An agent = something that sees + thinks + acts

Examples:

- A human (senses using eyes and ears, acts using hands and legs)

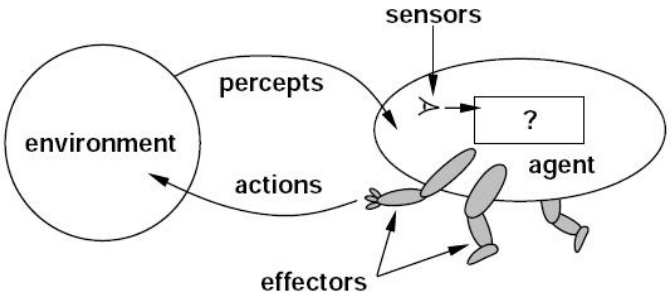
- A robot
- A software program like a chatbot

Intelligent Agent

An intelligent agent is an agent that:

- Observes the environment using sensors
- Performs actions using actuators
- Chooses actions intelligently to achieve its goal

An intelligent agent acts rationally, meaning it selects the best possible action based on what it knows.



Sensors and Actuators:

Every agent has:

- **Sensors** → to collect information from the environment
- **Actuators** → to perform actions in the environment

Agent	Sensors	Actuators
Human	Eyes, ears, skin	Hands, legs, mouth
Robot	Camera, infrared sensor	Motors, wheels, arms
Chatbot	Keyboard input	Text output
Self-driving car	Camera, radar, GPS	Steering, brakes

Example: Vacuum Cleaner Agent

A vacuum cleaner robot works as an intelligent agent.

- **Sensors:** Detect dirt and walls
- **Actuators:** Move left, move right, clean
- **Goal:** Clean all dirty areas

Behavior:

- If dirt is detected → it cleans
- If no dirt → it moves to another location

This simple sensing and acting behavior makes it an intelligent agent.

Environment in Artificial Intelligence

In Artificial Intelligence (AI), an environment is everything that surrounds the agent, where the agent exists, perceives, and takes actions.

Environment = The world around the agent

The agent interacts with the environment in two main ways:

- **Observing the environment** – sensing what is happening.
- **Acting on the environment** – performing actions that may change it.

Examples of Environment and Agents:

Example 1: Human as an Agent

- **Agent:** Human
- **Environment:** Room, road, society
- **Explanation:** Humans perceive their surroundings through senses and act within the environment to achieve goals.

Example 2: Vacuum Cleaner Robot

- **Agent:** Vacuum cleaner robot
- **Environment:** Rooms, dirt, walls
- **Explanation:** The robot senses the presence of dirt and performs cleaning actions to maintain a clean environment.

Agent–Environment Interaction:

The interaction between an agent and its environment is a continuous cycle.

Flow of Interaction:

Environment → Agent senses → Agent acts → Environment changes → Repeat

Explanation:

The **environment** provides input to the agent (through sensors).

The **agent** decides an action based on what it perceives.

The **action** of the agent affects the environment.

The changed environment is sensed again, and the cycle repeats.

Example:

A vacuum cleaner detects dirt → cleans it → dirt is removed → senses again if any dirt remains.

Types of Environments

In Artificial Intelligence, environments are classified into different types based on how an agent perceives the environment, how actions affect outcomes, and how the environment changes over time. This classification helps in designing appropriate intelligent agents.

1. Fully Observable Environment:

In a fully observable environment, the agent can perceive the complete state of the environment at any time through its sensors.

Example: Chess game — all pieces on the board are visible to the player.

2. Partially Observable Environment:

In a partially observable environment, the agent has limited or incomplete information about the environment.

Example: Driving a car — the driver cannot see all road conditions or other drivers' intentions.

3. Deterministic Environment:

An environment is deterministic when a particular action always leads to the same result.

Example: Calculator — pressing the same buttons always gives the same output.

4. Stochastic Environment:

An environment is stochastic when the same action may result in different outcomes due to uncertainty.

Example: Weather prediction — rain may or may not occur even with the same conditions.

5. Episodic Environment:

In an episodic environment, each decision is independent of previous decisions.

Example: Spam email filtering — each email is classified separately.

6. Sequential Environment:

In a sequential environment, current actions affect future states and decisions.

Example: Chess game — one move influences all future moves.

7. Static Environment:

A static environment does not change while the agent is making a decision.

Example: Crossword puzzle — the puzzle remains unchanged during solving.

8. Dynamic Environment:

A dynamic environment changes continuously while the agent is acting or thinking.

Example: Traffic system — vehicles move constantly and conditions change.

9. Discrete Environment:

A discrete environment has a finite number of states, actions, or perceptions.

Example: Chess — limited board positions and legal moves.

10. Continuous Environment:

A continuous environment has infinite or continuous states and actions.

Example: Robot movement — speed and position change continuously.

11. Single-Agent Environment:

In a single-agent environment, only one agent operates.

Example: Solving a puzzle alone.

12. Multi-Agent Environment:

In a multi-agent environment, multiple agents operate and interact.

Example: Road traffic — many drivers interact with each other.

PEAS Description of an Environment

Used to define agent tasks.

P – Performance measure

E – Environment

A – Actuators

S – Sensors

Example: Self-Driving Car

- **P:** Safety, speed, comfort
- **E:** Roads, signals, weather
- **A:** Steering, brakes, accelerator
- **S:** Cameras, radar, GPS

Nature of Environments [or] Properties of Environments

In Artificial Intelligence, the environment is the external world in which an intelligent agent operates. The nature of the environment describes the **characteristics of the environment** that affect how an agent perceives, decides, and acts. Understanding the nature of environments helps in designing intelligent agents that can behave effectively in different situations.

1. Observability of the Environment

The environment may be fully observable or partially observable depending on whether the agent can perceive the complete state of the environment.

- In a fully observable environment, the agent's sensors provide all relevant information needed to make decisions.

Example: Chess — all pieces are visible.

- In a partially observable environment, the agent has incomplete or uncertain information.

Example: Driving a car — the driver cannot see everything on the road.

2. Determinism of the Environment

The environment may be deterministic or stochastic based on how actions affect outcomes.

- In a deterministic environment, an action produces a fixed and predictable result.

Example: Calculator operations.

- In a stochastic environment, the same action can produce different results due to randomness or uncertainty.

Example: Weather conditions.

3. Episodic vs Sequential Nature

The environment may be episodic or sequential depending on whether actions depend on previous actions.

- In an episodic environment, each action is independent of previous actions.

Example: Image classification.

- In a sequential environment, current actions affect future decisions and states.

Example: Chess or driving.

4. Static vs Dynamic Nature

The environment may be static or dynamic depending on whether it changes while the agent is making decisions.

- In a static environment, the environment does not change during decision making.

Example: Crossword puzzle.

- In a dynamic environment, the environment changes continuously with time.

Example: Traffic system.

5. Discrete vs Continuous Nature

The environment may be discrete or continuous based on the number of possible states and actions.

- In a discrete environment, there are a limited number of states and actions.

Example: Board games.

- In a continuous environment, states and actions change continuously.

Example: Robot movement.

6. Single-Agent vs Multi-Agent Environment

The environment may be single-agent or multi-agent.

- In a single-agent environment, only one agent operates.

Example: Puzzle solving.

- In a multi-agent environment, multiple agents interact with each other.

Example: Road traffic system.

Concept of Rationality

Rationality means doing the right action at the right time. In Artificial Intelligence, rationality refers to how an intelligent agent chooses its actions.

A **rational agent** always selects the best possible action based on the information it currently has.

Rationality = choosing the best action with available knowledge

Rationality in Artificial Intelligence:

In AI, an agent may not know everything about the environment. Therefore, rationality does not mean perfection.

Important Point:

Rational ≠ Perfect

A rational agent:

- Does not need complete information
- Does not always produce the best outcome
- Only chooses the best possible action based on what it knows

Examples of Rational Behavior:

Example 1: Vacuum Cleaner Agent

- If there is **dirt → clean**
- If there is **no dirt → move to another location**

This behavior is rational because it helps the agent achieve its goal of keeping the room clean.

Example 2: Student as an Agent

- **Goal:** Get good marks
- **Action:** Study before the exam

Studying is a rational action because it increases the chances of achieving the goal. Not studying is an irrational action in this context.

Rationality of an agent depends on the following factors:

- **Percepts** – What the agent perceives from the environment
- **Knowledge** – What the agent knows about the environment
- **Actions** – What actions are available to the agent
- **Goals** – What the agent wants to achieve

An agent selects the action that best satisfies these factors.

Performance Measure:

Rationality is evaluated using a performance measure. A performance measure defines how successful an agent is in achieving its goal.

Examples:

- Vacuum cleaner → Cleanliness of the room
- Student → Exam marks
- Taxi driver → Safe, fast, and comfortable travel

A rational agent tries to maximize its performance measure.

Rational Agent:

An agent is called rational if:

- It performs the right action for the given situation
- It tries to maximize performance
- It uses available information effectively

Example:

If a vacuum cleaner cleans dirt when it detects dirt, it is acting rationally.

Structure of Agents

In Artificial Intelligence, an agent is an entity that perceives its environment and acts upon it. The structure of an agent explains how an agent is built internally and how it converts percepts into actions.

Basic Structure of an Agent:

An agent has three main components:

Agent = Sensors + Agent Program + Actuators

1. Sensors

Sensors are used by the agent to observe or sense the environment.

They collect information and send it to the agent.

Examples:

- Human → eyes, ears
- Robot → cameras, infrared sensors
- Software agent → keyboard input, data files

2. Agent Program

- The agent program is the brain of the agent.
- It decides what action to take based on the percepts received from sensors.
- It runs on a physical device called architecture
- It implements decision-making logic
- It maps **percepts → actions**

Example:

A chess program deciding the next move based on the current board position.

3. Actuators

Actuators are used by the agent to perform actions in the environment.

Examples:

- Human → hands, legs
- Robot → motors, wheels
- Software agent → screen output, file writing

Agent = Architecture + Program

1. **Architecture** → hardware or platform (computer, robot body)
2. **Program** → software that runs on the architecture

Together they form an intelligent agent.

Working of Agent Structure (Flow):

Environment → Sensors → Agent Program → Actuators → Environment

This cycle continues until the agent achieves its goal.

Example: Vacuum Cleaner Agent

- **Sensors:** Dirt sensor, location sensor
- **Agent Program:** Decides whether to move or clean
- **Actuators:** Wheels, vacuum motor

The agent senses dirt, decides an action, and cleans the area.

Importance of Agent Structure:

- Helps in designing intelligent systems
- Defines how perception leads to action
- Used to classify different types of agents

Types of Agent Structures [Or] Types of Intelligent Agents

An intelligent agent is any system that perceives its environment through sensors, makes decisions, and acts using actuators in order to achieve desired goals. The intelligence of an agent depends on how it decides what action to take. Based on this decision-making ability, intelligent agents are classified into different types.

1. Simple Reflex Agent

A simple reflex agent selects actions based only on the current percept. It uses a set of condition–action rules, which means: If a particular condition is observed, perform a specific action.

This agent does not consider the history of percepts and has no internal memory.

It reacts immediately to what it sees at the moment.

Example

A vacuum cleaner agent:

- If the room is dirty → Clean
- If the room is clean → Move to the next room

Characteristics

- Very fast and simple
- Easy to design
- Suitable only for fully observable environments

Limitations

- Fails in partially observable or complex environments
- Cannot improve or learn
- No understanding of the world beyond current perception

2. Model-Based Reflex Agent

A model-based reflex agent improves upon the simple reflex agent by maintaining an internal model of the environment. This model represents how the world works and how actions affect the environment.

The agent uses:

- Current percept
- Stored information from past percepts

This helps the agent deal with partially observable environments, where the agent cannot see the entire environment at once.

Example

A smart vacuum cleaner:

- Remembers which rooms are already cleaned
- Avoids unnecessary cleaning

Characteristics

- Has memory
- Can track changes in the environment
- More reliable than simple reflex agents

Limitations

- Still rule-based
- No planning or goal reasoning

3. Goal-Based Agent

A goal-based agent selects actions by considering future states and checking whether those states help in achieving a goal. This requires search and planning.

Example

Route-finding agent:

- Goal: Reach destination city
- Chooses roads that lead toward the destination

Characteristics

- Flexible behavior
- Can choose different paths to reach the goal
- Suitable for problem-solving tasks

Limitations

- Slower than reflex agents
- Requires more computation

4. Utility-Based Agent

A utility-based agent goes beyond goal achievement by selecting actions that maximize utility, which represents how desirable an outcome is. When multiple solutions achieve the goal, this agent chooses the best one.

Example

Self-driving car:

- Chooses a route that is safe, fast, and comfortable
- Avoids traffic even if the route is longer

Characteristics

- Can handle trade-offs
- Works well in uncertain environments

- Makes rational decisions

Limitations

- Designing utility functions is difficult
- Computationally expensive

5. Learning Agent

A learning agent improves its performance by learning from experience. It can adapt to changes in the environment and improve over time.

Main Components

1. **Performance element** – chooses actions
2. **Learning element** – improves performance
3. **Critic** – provides feedback
4. **Problem generator** – explores new actions

Example

- Game-playing AI
- Recommendation systems

Characteristics

- Self-improving
- Adapts to new situations
- Most advanced type of agent

Relationship Between Agent Types

Agent intelligence increases in the following order:

Simple Reflex → Model-Based → Goal-Based → Utility-Based → Learning Agent

Each new type adds more reasoning ability.

Agent Type	Memory	Goal	Utility	Learning
Simple Reflex	No	No	No	No
Model-Based	Yes	No	No	No
Goal-Based	Yes	Yes	No	No
Utility-Based	Yes	Yes	Yes	No
Learning Agent	Yes	Yes	Yes	Yes

Problem-Solving Agents

A problem-solving agent is an intelligent agent that decides what actions to take by searching for a sequence of actions that will take it from its current situation to a desired goal.

Instead of reacting immediately to what it senses, a problem-solving agent plans before it acts. It first thinks about the problem, finds a solution, and then follows that solution step by step.

Basic Idea of a Problem-Solving Agent

The main idea behind a problem-solving agent is **"think first, act later."**

The agent:

- Understands where it is now
- Decides what it wants to achieve
- Plans a sequence of actions
- Executes the plan

This makes problem-solving agents different from simple reflex agents, which act without planning.

Steps Followed by a Problem-Solving Agent:

A problem-solving agent works in a fixed sequence of steps.

1. Perceiving the Environment

The agent first uses its sensors to observe the environment and identify its current state.

Example:

A robot senses that it is in Room A and that the room is dirty.

2. Goal Formulation

After understanding the current situation, the agent decides what it wants to achieve. This desired outcome is called the goal.

Example:

The goal is to clean all rooms.

3. Problem Formulation

The agent then defines the problem in a formal way so that it can be solved.

This includes:

- **Initial state** – the starting situation
- **Actions** – what the agent can do
- **Goal state** – the condition to be achieved

Example:

Initial state: Robot in Room A

Actions: Move left, move right, clean

Goal state: All rooms are clean

4. Searching for a Solution

The agent now searches through possible sequences of actions to find one that leads from the initial state to the goal state. This sequence of actions is called a solution.

Example:

Clean Room A → Move to Room B → Clean Room B

5. Executing the Solution

Once a solution is found, the agent executes the actions one by one in the real environment. The agent does not search again unless the environment changes.

Important Characteristics of Problem-Solving Agents:

- They work in fully observable environments
- The environment is usually static while planning
- Actions have predictable results
- Problems are well-defined
- They rely on search and planning

Because of these assumptions, problem-solving agents are suitable for simple and structured problems.

Example: Route-Finding Agent

A route-finding agent plans how to travel from one place to another.

- Initial state: Starting city
- Goal state: Destination city
- Actions: Travel between connected cities
- Solution: A sequence of roads leading to the destination

The agent searches a map and chooses the best route.

Problem Formulation

Problem formulation is the step in which an intelligent agent clearly defines a problem before trying to solve it.

In simple words, problem formulation means: Telling the agent where it is, what it wants to achieve, and what actions it can use to reach the goal. Without problem formulation, an agent cannot think or plan.

Importance or Need of Problem Formulation:

Problem formulation is important because:

- It helps the agent understand the problem clearly
- It tells the agent which actions are allowed
- It defines when the goal is achieved
- It helps in finding a correct and efficient solution

If the problem is not formulated properly, the agent may:

- Choose wrong actions
- Never reach the goal
- Waste time searching

Main Elements of Problem Formulation:

A problem is defined using five basic components.

1. Initial State

The initial state is the starting situation of the agent.

Example:

A vacuum cleaner is in Room A, and the room is dirty.

2. State Space

The state space is the set of all possible situations the agent can be in.

It includes:

- The starting state
- All possible intermediate states
- All possible goal states

Example:

Vacuum in Room A or Room B

Each room may be clean or dirty

3. Actions (Successor Function)

Actions are the moves or operations the agent can perform.

The successor function tells:

- What actions are possible
- What new state results after each action

Example (Vacuum Cleaner):

- Move left
- Move right
- Clean

4. Goal Test

The goal test checks whether the agent has achieved its goal.

Example:

All rooms are clean.

5. Path Cost

The path cost measures how expensive a solution is.

It may represent:

- Number of steps
- Time taken
- Energy used

The agent tries to find the solution with the minimum cost.

Example:

Each action costs 1 unit → total cost = number of actions.